Samuel Agyapong

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SUMMARY

Electronics Engineering Technology student with hands-on experience in digital logic design, circuit-level hardware development, PCB design, and structured hardware validation. Strong foundation in Verilog HDL, developed through FPGA-based digital systems projects involving FSM architecture, timing design, and testbench-driven verification. Proficient in KiCad (PCB design), Verilog HDL, Python, Embedded C, AutoCAD, LTspice, Multisim, and MATLAB; experienced with oscilloscopes and multimeters for circuit debugging and signal analysis. Eager to contribute to hardware design, testing, and validation work on cutting-edge computing platforms. CPT authorized – no sponsorship required.

EDUCATION

Grambling State University | Grambling, LA — Aug 2024 - May 2028

Bachelor of Science, Electronics Engineering | GPA: 3.88 / 4.0 | President's List (Spring 2025)

Relevant Coursework: Circuit Analysis I & II • Electronic Devices • Engineering Materials • Calculus I & II • Computer Applications (Python)

TECHNICAL SKILLS

- **Hardware:** Arduino, Breadboard Prototyping, Circuit Design & Debugging, Multimeter & Oscilloscope,
- **Software:** Python, Embedded C, Verilog HDL, SolidWorks, AutoCAD, KiCad (PCB Design), Multisim, LTspice, MATLAB, MS Office
- **Tools & Concepts:** Digital Logic, Microcontrollers, Sensor Integration, Signal Measurement, Soldering

RELEVANT PROJECTS

Line-Following Robot | Arduino, Infrared Sensors — Team Project — Dec 2024

- Designed and built a custom autonomous hardware platform integrating infrared sensors, motor drivers, and Arduino-based control logic; handled full circuit-level integration from breadboard layout through iterative debugging to a verified, functional system.
- Used oscilloscope and multimeter measurements to diagnose signal-level failures during hardware testing; resolved sensor calibration issues through targeted circuit-level adjustments rather than full redesign.

Lightweight Drone Frame Design | AutoCAD — Intro to Drone Technology — May 2025

- Designed a modular UAV airframe in AutoCAD optimized for structural efficiency; applied static load analysis and 3D simulation to validate design performance before fabrication — practicing the same pre-silicon validation mindset used in hardware product development.

DWIPS — Wearable Motion-Detection Safety System | Sensors, Embedded Logic, System Integration — Team Project — Oct 2025

- Designed and validated a wearable safety system integrating motion-detection sensors with embedded control logic and a custom PCB-based alarm circuit; led sensor integration, circuit-level debugging, and iterative hardware validation across 4-person team to verify system reliability under real-world conditions; presented at a competitive pitch event earning 3rd place.

Image Processing System for Object Detection | Python, OpenCV — Jan 2026

- Built a real-time object detection pipeline in Python and OpenCV to classify electronic circuit board components — a computer vision task directly relevant to hardware inspection workflows used in high-volume electronics manufacturing and production environments.
- Implemented preprocessing stages including grayscale conversion, Gaussian blur, and adaptive thresholding; validated detection reliability across multiple component categories through structured testing, documenting failure modes and performance at each iteration.

FPGA Traffic Light Controller | Verilog HDL | Icarus Verilog | GTKWave — Mar 2026

- Designed a Moore FSM-based traffic light controller in Verilog HDL implementing 4-state transition logic (RED, GREEN, YELLOW, PEDESTRIAN) with parameterized timing for FPGA deployment on Xilinx Basys 3 via Vivado synthesis.
- Wrote a structured testbench in Icarus Verilog to verify all FSM state transitions, counter logic, and edge cases; analyzed correctness via GTKWave waveform inspection — mirroring simulation-based verification methodology used in ASIC and FPGA hardware validation.
- Extended the design with pedestrian request latching and dual-intersection anti-phase sequencing — demonstrating the ability to layer increasing logical complexity onto a verified hardware base without breaking prior functionality.
- Designed a 4-bit countdown timer with BCD seven-segment display decoding output, implementing the full digital signal path from internal counter logic to external hardware interface — practicing board-level signal design directly relevant to hardware product development.
- Structured the full project with modular RTL design, separating FSM logic, counter modules, and display decoding into independently testable units; maintained a version-controlled GitHub repository with documented commits following hardware development best practices.

RESEARCH & LEADERSHIP

Perovskite Solar Cell Characterization & Reliability Research | Grambling State University — Spring 2026 – Present

- Conduct collaborative research on perovskite solar cells, integrating graphene-based carbon quantum dots into active layers to improve device stability and performance.
- Perform solar cell characterization using an Ossila solar simulator, including current–voltage (J–V) analysis under standard illumination to benchmark prototype cells against commercial devices.
- Support material and reliability analysis using optical spectroscopy, photoluminescence, and SEM while contributing to experimental planning, data collection, and technical documentation.

Student Assistant — Drone Technology & Electronic Devices (High School Day) | Grambling State University — Spring 2025 – Present

- Led STEM outreach demonstrations on UAV systems and core EE concepts using hands-on systems including a DC–AC converter, PCB-based alarm circuit, and hand-crank generator.
- Translated complex technical ideas into clear explanations for non-technical audiences; contributed to recruitment of ~2–3 participants who later enrolled at the university.